

GLS UNIVERSITY
Faculty of Computer Applications & Information Technology

Integrated MSc(IT)

221601606 DISSERTATION / MINI PROJECT

(Skill Enhancement Course)

Title Approval Form – Mini Project

A. Student's Details:

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B. Company Detail:

Company Name	Email ID	Contact Number
Markoryx solution	faizan@markoryxsolution.com	+91 6354923686

Company Link : www.markoryxsolution.com

C. Project Detail:

Project Title:	TeachTwin Ai
Front-end Technology:	HTML, CSS, JS
Back-end Technology:	MongoDB, Node.JS

★ **Project Description:**

An AI-powered assistant that replicates the exact voice, tone, and teaching style of each subject teacher. After the lecture is over, students can interact with this bot for doubts, revision, or even assignment guidance — as if they were talking to their real teacher.

D. Functionalities:

- Core Concept

- Each subject has a dedicated teacher.
- We capture their **voice, tone, way of explaining**, and **content**.
- Students can talk to the teacher's AI replica anytime — even late at night.
- Bot gives replies in the **teacher's real voice** (AI cloned).
- Keeps learning from the teacher's new lectures, notes, and assignments.

- Main Point Solved

Problem	Solution
Students forget what was taught	Bot recalls the exact lecture instantly
Students hesitate to ask doubts	Ask anytime without fear or shame
Teachers can't be available 24x7	AI handles follow-ups post class
Every teacher has different way of teaching	AI maintains each teacher's identity

- High-Impact Features

- **Teacher Voice Cloning:** Feels like the real teacher is answering.
- **Per-Subject Memory:** Keeps a contextual memory of every subject's content.
- **24/7 Access:** Students can ask questions any time.
- **Assignment Integration:** Understands and explains teacher’s given homework.
- **Self-Updating AI:** Automatically updates itself with teacher’s new material.
- **Emotionally Familiar Replies:** Makes students feel comfortable and connected.

- Technical Overview

- **Frontend:** Simple chat interface (Web + App)
- **Voice AI:** Teacher’s voice cloned using free tools (e.g., OpenVoice, Bark, Coqui)
- **Database:** Stores subject-wise content, tone profiles, and voice embeddings
- **Backend AI:** Powered by GPT-4 (via API), prompt-engineered with teacher context
- **Storage:** Free Firebase, Supabase, or SQLite Lite-tier
- **Hosting:** Vercel / Netlify (frontend), Render / Railway (backend)

- Why It Feels Like a Cheat Code

- Students are emotionally attached to their teacher's voice
- Teacher is digitally present even after class ends
- Teachers' efforts get **digitally multiplied**
- No extra work for the teacher

- Proof of Value

- “80% of students hesitate to ask doubts in class” — solved.
- “Teachers can’t repeat every concept 10x” — AI will.
- “Students feel more connected to a voice they know” — solved.

- Ideal for:

- Schools
- Coaching centers
- Online course creators
- Remote learning platforms

- Sample Use Case:

"A Class 9th student forgot the concept of photosynthesis. It's 10 PM. He types a question. The bot replies in the **exact tone and voice** of his Bio teacher. Feels like ma'am is speaking to him directly."

- Summary:

You're not just giving a chatbot. You're giving **teachers their digital twin**. And **students their favorite teacher on-demand**.

- Future Upgrade:

- Students can reply with **voice**, and get reply in **teacher's voice** (2-way audio)
- Personal feedback & motivation voice drops from teacher AI

Modules:

-- Admin

- ➤ User and Organization Management
- ➤ Content Management
- ➤ Reporting and Analytics
- ➤ Feedback and Support Management

-- Users (Students)

- ➤ User Registration and Profile Management
- ➤ Lecture Recall and Subject-wise Memory
- ➤ Doubt Clearing & Assignment Guidance

- ➤ Notifications and Alerts
- ➤ Review and Rating for teacher AI experience

-- Teachers

- ➤ Teacher Profile & Digital Identity Setup
- ➤ Lecture & Content Upload Management
- ➤ Student Interaction Analytics
- ➤ Volunteer/Support Opportunity Management
- ➤ Communication with Students via AI Assistant